Senthil Nathan

Lead Machine Learning & MLOps Engineer

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 5+ years of production experience designing statistical ML models, computer vision systems, and robust MLOps pipelines. Proficient in PyTorch, Scikit-Learn, MLflow model tracking, Kubeflow orchestration, and model quantization (ONNX, TensorRT).

CORE TECHNICAL STACK

ML & Deep Learning
PyTorch, TensorFlow, Scikit-Learn, XGBoost, LightGBM, OpenCV, TorchVision

MLOps & Deployment
MLflow, Kubeflow, Feast Feature Store, Triton, ONNX Runtime, Docker, K8s

Data Engineering
Python, SQL, Apache Spark (PySpark), DVC, AWS SageMaker, GCP Vertex AI

WORK EXPERIENCE

Lead Machine Learning Engineer • **Tiger Analytics**
Aug 2021 – Present | Chennai

- Built predictive churn models with XGBoost on PySpark pipelines across 50M customer rows, increasing F1-score to 0.89.
- Implemented Kubeflow pipelines and MLflow model registry, shortening model re-training cycles from weeks to 2 hours.
- Quantized CNN inference pipelines via TensorRT, cutting edge server inference latency from 140ms to 24ms.

Machine Learning Engineer • **Robert Bosch India**
Jul 2019 – Jul 2021 | Coimbatore

- Trained YOLOv5 computer vision defect detection models for automotive manufacturing quality lines with 98.6% mAP.
- Constructed real-time feature extraction pipelines leveraging Redis and Kafka for low-latency scoring.
- Authored unit and regression tests for data drift using Great Expectations and Evidently AI.

FEATURED ML PROJECTS

Automated Credit Risk & Default Prediction System
Developed an ensemble gradient boosting classifier incorporating SHAP explainability values for transparent regulatory compliance.

Real-Time Edge Computer Vision Pipeline
Implemented lightweight MobileNetV3 visual inspection models compiled with OpenVINO deployed on edge industrial gateways.

EDUCATION & CREDENTIALS

B.E. Electronics & Communication • Government College of Technology (GCT), Coimbatore

TensorFlow Developer Certified • AWS Certified Machine Learning - Specialty